# Expertise Validation Form

Automated Viva Assessment Using Multi-Modal Analysis (Video, Audio and Speech), Live Interviewer Copilot, and Large Language Models.

## Academic Expert Review and System Validation

### 1. Expert and Evaluation Details

**Lecturer's Full Name:** .....

**Designation / Academic Position:** .....

**Faculty / Department:** .....

**Institution:** Sri Lanka Institute of Information Technology (SLIIT)

**Date of Evaluation:** .....

### 2. Student Details and Research Details

**Student Name –** B.K.H.M.B.L. Herath

**Student Id –** IT22557056

**Research Group Id –** R26-SE-025

**Research Title -** AI-Integrated Multi-Modal Academic Evaluation, Performance Intelligence, and Automated Viva Assessment System

### 3. Purpose of Validation

This document is intended to obtain an independent expert's assessment of the Viva Evaluation System developed for academic assessment, comprising (a) analysis of recorded viva videos and (b) the Live Interviewer Copilot used during live viva sessions. The expert is requested to review the system demonstration and/or sample test cases and evaluate its functionality, accuracy, marking consistency, and suitability for academic use.

Please tick **one option** for each criterion based on the evidence demonstrated. Select "Cannot Validate" if the criterion cannot be confirmed or requires further testing. Additional comments and recommendations are welcome.

### 4. Scoring Metric Used by the System

The official AI performance mark (/100) is the equal-weighted average of the available components below. A component with no usable data is dropped and the rest are renormalised. Facial emotion, speech emotion, gaze, head pose and blink are reported as supporting evidence and are not part of the official mark; they are not yet validated against labelled viva data.

| Component | Metrics | Role in mark |
|---|---|---|
| Engagement | CNN frame engagement score; facial confidence | Scored (1/3) |
| Audio acoustics | Pitch stability, HNR clarity, jitter and shimmer | Scored (1/3) |
| Transcript | Speech rate band, hedges, fillers, long pauses, sentence completion | Scored (1/3) |
| Facial emotion / gaze / head / blink | Emotion profile, on-camera gaze ratio, head stability, blink rate | Evidence only |
| Speech emotion | Emotion label and confidence from voice | Evidence only |

| Component | Metrics | Role in mark |
|---|---|---|
| Technical accuracy (optional) | Examiner-confirmed score /10, final = 50% AI + 50% technical | Scored when enabled |

**5. System Validation Checklist**

| No. | Validation Criterion | Can Validate | Cannot Validate |
|---|---|---|---|
| 1 | The system accepts a recorded viva video (upload) or a live viva session and processes it successfully. | ■ | ■ |
| 2 | The system detects the student's face and applies a quality gate; recordings without a visible face are flagged as incomplete rather than marked. | ■ | ■ |
| 3 | The system recognises facial expression/emotion from the video and reports an emotion profile over time. | ■ | ■ |
| 4 | The system computes a frame-level engagement score from the student's facial behaviour (CNN engagement model). | ■ | ■ |
| 5 | The system measures eye gaze (on-camera ratio), head stability and blink rate as supporting behavioural evidence. | ■ | ■ |
| 6 | The system converts the student's speech to text accurately (audio extraction and speech-to-text). | ■ | ■ |
| 7 | The system recognises emotion from the student's voice (speech emotion recognition) and reports it as supporting evidence. | ■ | ■ |
| 8 | The system measures voice quality from the audio (pitch stability, clarity/HNR, jitter and shimmer). | ■ | ■ |
| 9 | The system analyses speech delivery from the transcript (speech rate, long pauses, filler words, hedging, sentence completion). | ■ | ■ |
| 10 | The system evaluates the relevance of the student's answers to the examiner's questions (Q&A relevance). | ■ | ■ |
| 11 | The system supports technical-accuracy assessment against the subject rubric or project context, with the examiner confirming the technical score. | ■ | ■ |
| 12 | The scoring metric in Section 4 is applied correctly: the official mark (/100) is computed from the weighted components, and unavailable components are excluded rather than penalised. | ■ | ■ |
| 13 | The system shows a per-component score breakdown so the examiner can see how each metric contributed to the final mark. | ■ | ■ |
| 14 | The system supports the examiner workflow for marks (draft, review and publish); technical-accuracy marks are never published automatically. | ■ | ■ |
| 15 | The Live Interviewer Copilot transcribes the student's speech in real time during the presentation and viva. | ■ | ■ |

| No. | Validation Criterion | Can Validate | Cannot Validate |
|---|---|---|---|
| 16 | The Live Interviewer Copilot suggests relevant follow-up questions, and the examiner decides which to ask (human-in-the-loop). | ■ | ■ |
| 17 | The evaluation results and AI-generated feedback are understandable and useful. | ■ | ■ |
| 18 | The system produces reasonably consistent results for comparable recordings. | ■ | ■ |
| 19 | The system's workflow and outputs are suitable for the intended academic users (lecturers and examiners). | ■ | ■ |
| 20 | Overall, the system demonstrates potential for supporting viva assessments in academic settings. | ■ | ■ |

### 6. Overall Expert Assessment

Based on the system demonstration and evidence reviewed, please select one option.

[ ] **Can Validate:** The system is suitable for the intended viva evaluation task.

[ ] **Can Validate with Improvements:** The system is suitable after the recommended improvements are completed.

[ ] **Cannot Validate at Present:** The system requires further development and/or testing.

[ ] **Unable to Determine:** There is insufficient evidence to make an overall assessment.

### 7. Expert Comments and Recommendations

**Strengths identified:**

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**Issues identified / Recommended improvements:**

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### 8. Expert Confirmation and Signature

I confirm that I have reviewed the Viva Evaluation System through a demonstration and/or sample test case. The selections and comments recorded in this document represent my professional assessment based on the evidence presented.

**Expert's Full Name:** .....

**Designation:** .....

**Signature:** .....

**Date:** .....

Thank you for your valuable expert contribution to this research and system validation process.